Method Statement

Ref. #: DCP00/08-0197-A-2025



DonProof R

(High quality FPO/TPO waterproofing membrane)

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Section A: General Comments

General Notes:

The information below is a detailed overview of the application of DCP's **DonProof R** waterproofing membrane and should be read in conjunction with the relevant technical data sheet prior to application. All DCP Products should be applied by experienced specialist contractors.

All the points below assume the correct preparation of the relevant surface.

Tools and Equipment:

It is suggested that the following list of equipment are adopted as a minimum requirement

Personal protection	:	*Protective overalls*
	:	*Goggles or a face mask*
	:	*Good quality gloves*
	:	*Safety shoes*
	:	*Safety helmet*
Application equipment	:	*Manual hot-air blower (Fig.1)*
	:	*Automatic hot wedge welder (Fig.2)*
	:	*Utility knife with blade (Fig.3)*
	:	*Pressure roller (Fig.4)*
	:	*Nail Gun + Nails (Fig.5)*





Fig.1: Manual hot-air blower



Fig.2: Automatic hot wedge welder



Fig.3: Utility knife with blade



Fig.4: Pressure roller



Fig.5: Nail Gun + Nails



Fig.7: Measuring & marking tools

Section B: Application

1.0 Substrate Preparation

1.1 Roof areas must have a continuous slope leading toward water outlets, following flat roof regulations.

1.2 A slope of at least 1:40 is recommended, but local or national regulations must be followed.

1.3 Water outlets should always be installed at the lowest points of the roof.

Steel Decking	• Only galvanized steel sheets with factory-applied corrosion protection should be used. • Minimum sheet thickness should be 0.75 mm, or as specified by the fastener manufacturer or local regulations.
Concrete	• Concrete roofs must be properly cured and dry as per flat roof standards or fastener manufacturer instructions. • The surface should be smooth, even, and clean, free from rocks, cracks, sharp edges or projections, gratings and nibs
Pre-Cast Concrete Elements	• Installed elements must form a plain and even surface. • Construction joints must be sealed and closed. • Supporting joints (butt ends) should be covered with protective strips mechanically fastened to prevent movement.
Timber Decking and Wooden Derivatives	• The wooden structure must be protected against moisture. • Board thickness and support spacing should match the expected load: ▪ **Timber boards**: minimum 24 mm thick ▪ **Plywood/OSB**: minimum 18 mm thick • Follow local or national standards, which may vary by product type, quality, thickness, and fasteners used.
Old Roof Surfaces (Renovation)	• Before renovation, open several areas down to the load-bearing layer to check: ▪ Vapour barrier condition ▪ Thermal insulation thickness and moisture level ▪ Possibility of drying the roof if moisture is found • On bitumen substrates, remove waves, blisters, and other uneven areas mechanically, and level the surface. • In some cases, a protective layer may be required.
Vapour Control Layers	• Install a continuous vapour control layer to prevent moisture migration and heat loss. • Ensure all laps and terminations are taped and sealed for airtightness. • In compact/warm roofs, polyethylene (PE-foil) is usually sufficient, but requirements depend on the building type. Buildings with high humidity (e.g., swimming pools) need higher-performance vapour control layers. • If unsure, consult the client about the vapour control layer's suitability and quality.

2.0 Separation/Protection Layers below the Membrane

2.1 Protection layers are primarily used to separate incompatible products or to prevent damage.

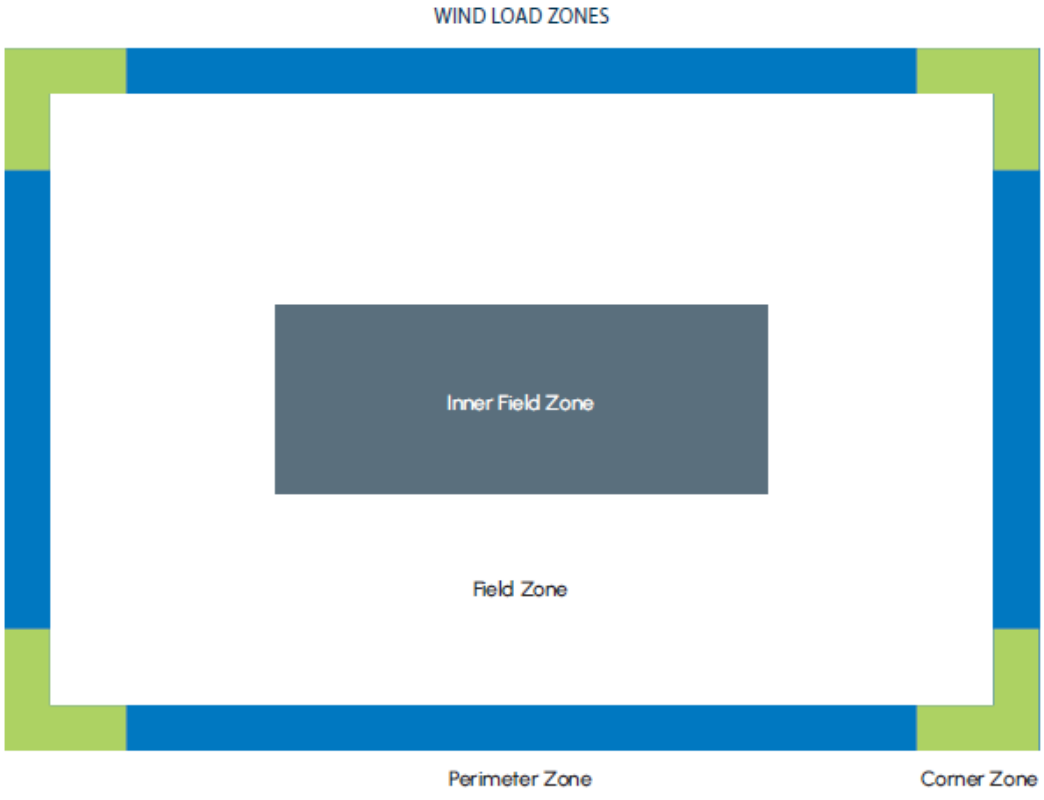
2.2 **DonProof R** membrane does not require any separating layers when in contact with bitumen and polystyrene as they are fully compatible.

2.3 Protective layers are mandatory over concrete and recommended over other substrates to prevent damage from rough or uneven surfaces, especially bituminous ones to avoid discoloration.

2.4 Suitable materials include polyester or polypropylene fleeces (min. 300 - 350 g/m²), or a single layer of insulation (PIR/PUR, Mineral wool (stone wool/glass wool), XPS or EPS polystyrene) applied over a vapor control layer.

2.5 In protected systems, If the ballast consists of crushed stone with sharp edges, the roofing membrane must be protected with polyester or polypropylene fleece or similar with a minimum surface weight of 300 g/m².

2.6 Prior to placement, surfaces should be thoroughly cleaned to ensure they are free from any loose materials, sharp projections, or debris that could damage the membrane.



3.0 Application of DonProof R Membrane

3.1 Exposed Roof Systems (Mechanically Fixed)



3.1.1 Roll out **DonProof R** membrane in the desired direction.

3.1.2 Straighten and tighten the sheet to eliminate wrinkles and ensure it lies flat and fix the ends to keep the membrane in place.

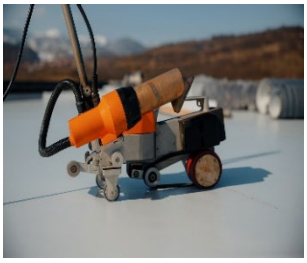


3.1.3 Fix the membrane mechanically along one longitudinal edge with fasteners placed 30 mm from the edge of the sheet. This anchoring provides stability for welding and prevents movement.

3.1.4 Lay the next sheet with a minimum overlap of 120 mm (Use the pre-marked guideline located 110 mm from the edge to ensure accurate alignment and consistent overlap.)

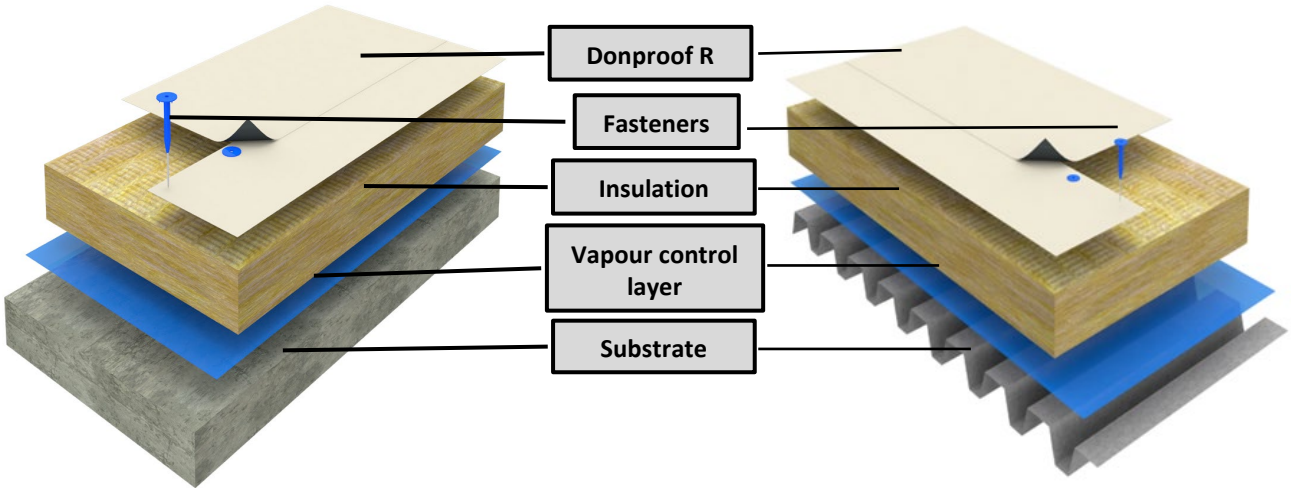
3.1.5 Weld the lap using an automatic hot-air machine (temperature 400 - 600°C, nozzle width 40 mm).







3.1.6 Maintain ≥ 10 mm distance from washer plate to membrane edge.



3.2 Protected Roof Systems (Ballasted/Green Roofs)

3.2.1 No mechanical fixing is required, the ballast provides wind protection. Minimum overlap ≥ 80 mm.

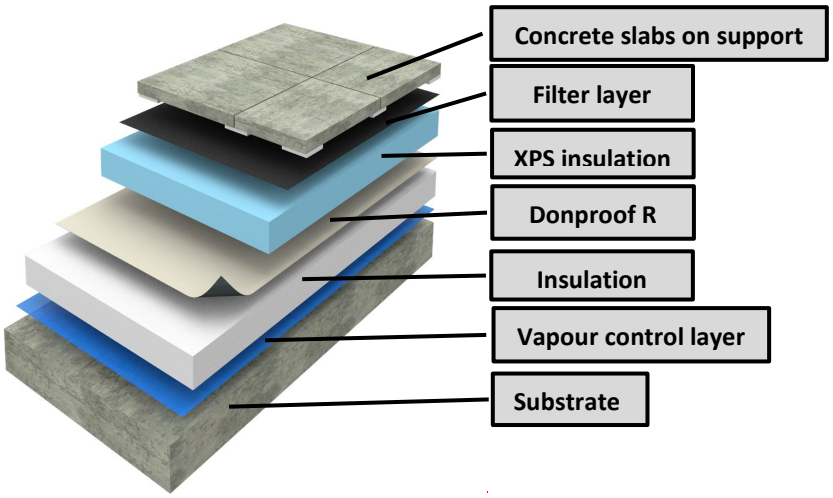
3.2.2 Apply geotextile and ballast (rounded pebbles or pavers) after welding.

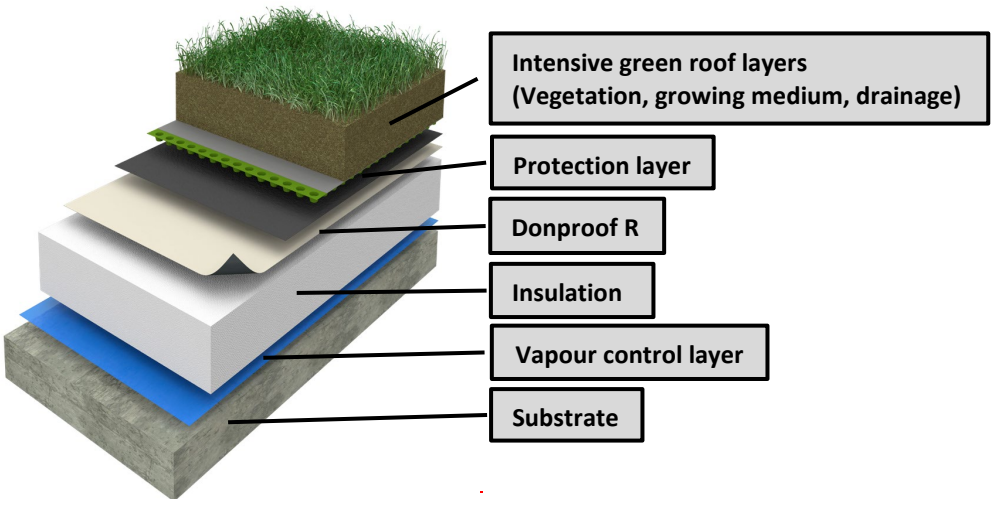
3.2.3 The ballast layer is used to stop wind from lifting or damaging the roofing membrane. The type and weight of the ballast are both important — lighter materials like fine gravel can blow away more easily than heavier ones such as concrete slabs.

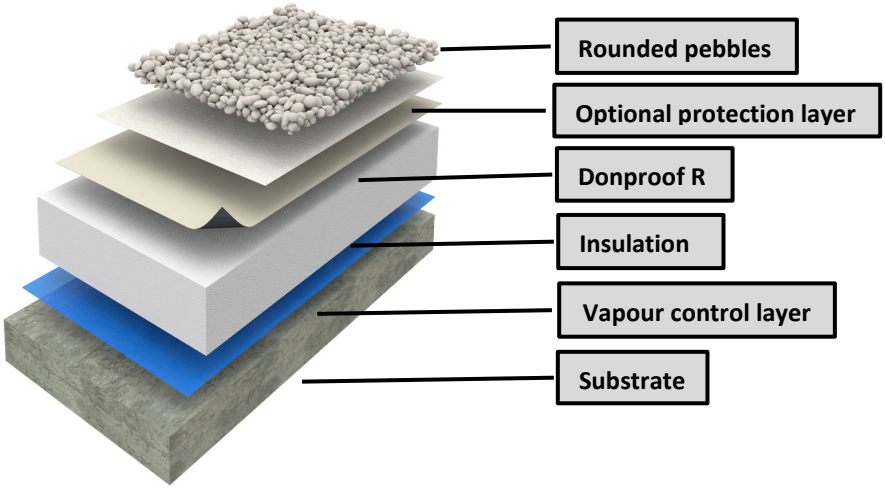
3.2.4 For airtight roofs, apply a 50 mm layer of round-grained natural gravel with a size range of 16 - 23 mm. This provides good protection against normal wind loads, especially in exposed roof corners, when the design load (Qd) is less than 3 kN/m².

3.2.5 Where stronger wind resistance is needed, use gravel graded 16 - 32 mm, which can withstand wind speeds up to 80 m/s before moving. The design wind speed should follow the relevant building wind load standards.

3.2.6 If required, 50 mm concrete tiles can also be used as ballast to protect the roof from continuous wind suction up to Qd ≤ 5 kN/m².







4.0 Protective Layers above the Membrane

4.1 A protective layer is required above the membrane in ballasted systems, especially if the roof will be used as a recreational area or if gravel is applied using pneumatic blowing.

4.2 In green roofs, the drainage or water retention layers typically serve this protective function.

5.0 Detailing of Overlaps

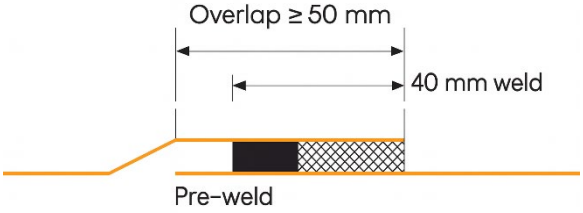
5.1 **Pre-weld:**

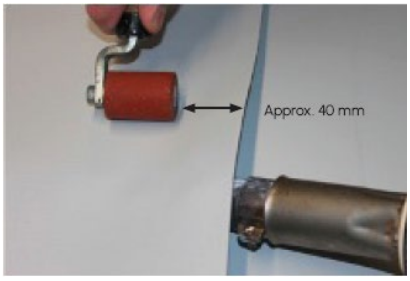
5.1.1 Begin by creating a pre-weld line.

5.1.2 Fix the upper membrane sheet approximately 50 mm from the edge by pressing the silicone roller down onto the edge of the underlying sheet.

5.1.3 This creates a narrow initial weld that prevents hot air from escaping during the main welding process and helps maintain consistent temperature.

5.1.4 The pre-weld also ensures proper sheet alignment and positioning for final welding.



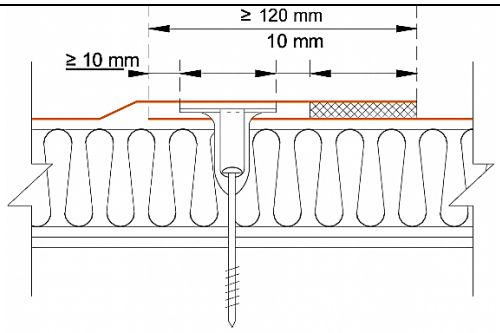
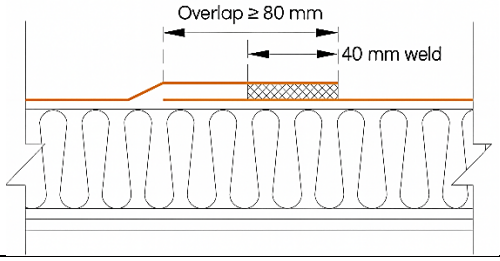
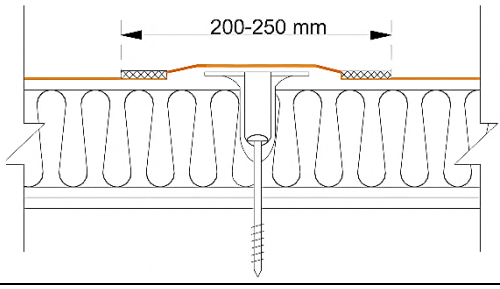


5.2 Main Weld

5.2.1 Perform the main weld over the remaining 40 mm width of the overlap. Move the hot air welder and silicone roller in a slow, continuous motion along the seam. This controlled movement ensures consistent heating and pressure, resulting in a watertight and durable weld.

5.2.2 Once welding is complete, reduce the temperature setting on the tool, but allow the blower to continue running until the air has cooled. This step protects the heating elements and extends the life of the tool.

The overlap consists of:

Exposed systems: • ≥ 10 mm distance from the washer plate to the membrane edge, • 40 - 50 mm width washer plate, • 10 mm tolerance and approximately 40 mm hot air welding.	
Protected systems: • Membrane is wind protected by the ballast thus no fasteners are required. • The overlap can be reduced from 120 mm to ≥ 80 mm.	
Symmetrical Fastening System with intermediate fastening rows: This system is mainly used as an intermediate fixing method in perimeter and corner areas instead of using narrower membrane widths. The intermediate fixing lines are covered by **Donproof R** membrane strips, width approximately 200 - 250mm.	

5.3 **Connection of new membranes to existing membranes**

5.3.1 New membrane sheets can be welded to **previously installed membranes**.

5.3.2 Before welding, the surface of the existing membrane must be thoroughly cleaned using a one-handed angle grinder fitted with a rotating wire brush to remove all dirt, oxidation, and patina layers.

5.4 **T-Joints**

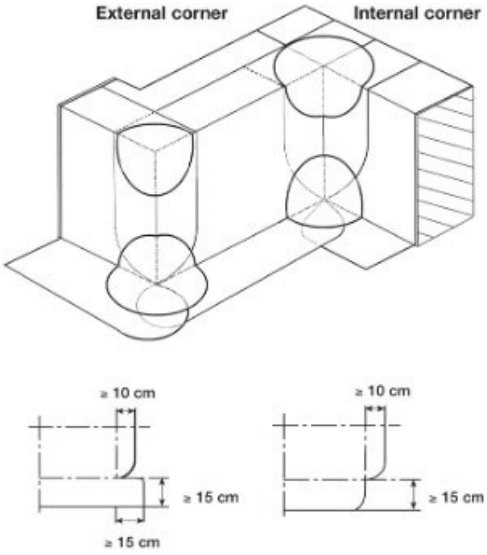
5.4.1 T-joints are used where three membrane sheets intersect. It is recommended to use an edge trimmer to cut the overlapping section between the two lower membrane sheets at the intersection point.

5.4.2 Trimming the overlap edge makes installation faster and easier, while also helping to achieve a watertight weld at the T-joint.

5.4.3 All T-joints must be reinforced with patches, which can be either site-fabricated or prefabricated.

5.5 **Internal and external corner**

5.6 Homogeneous membrane strips are used and welded around corner areas for waterproofing.



5.6.1 Sealing can be done using
- Prefabricated external or internal corners, or
- Custom round corner protections made from homogeneous membrane.

5.6.2 The size and shape of the corner protection depend on the specific corner detail being sealed.

5.6.3 External corners are also suitable for sealing right-angled or square roof penetrations, such as ventilation pipes and skylights (dome lights).

6.0 Welding Seam Control



6.1 Before beginning installation, test welding must be performed to establish the correct welding parameters and confirm acceptable weld quality.

6.2 Seam quality should be checked using probe test and verified through a peel test. These tests should only be carried out after the weld has fully cooled—either 24 hours after welding or following immersion in cold water.

6.3 The membrane temperature during testing must be below 20°C.

6.4 During installation, a test weld must be completed at the start of each machine-welding session. In addition, peel tests should be performed at intervals of approximately every 200 linear meters to confirm ongoing welding quality.

7.0 Details and Termination Work

7.1 Flashings

7.1.1 Flashings are made using steel strips mechanically fastened to resist horizontal forces, fixed at ≤ 200 mm centers.

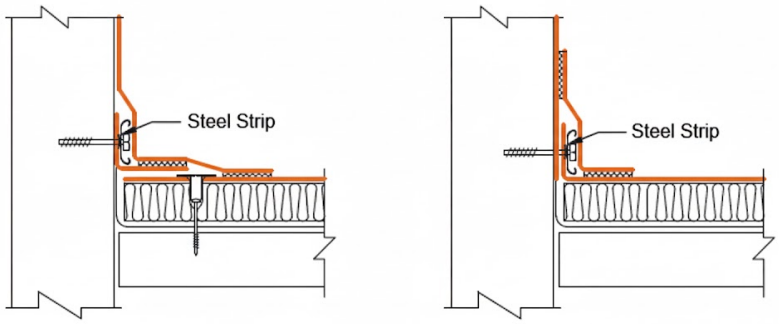
7.1.2 Details must be windproof and designed to prevent water from running behind or under the membrane.

7.1.3 The flashing-to-field membrane connection must be no more than 20 cm from the upstand.

7.1.4 Sealants or mastics may be used to seal profiles.

7.1.5 Perimeter fixings (around upstands and penetrations) require approximately 5 fasteners per linear metre using steel bars or laminated metal strips.

7.1.6 In valleys with a slope change of ≥ 3°, and at wall connections, install fixings at ≤ 200 mm.

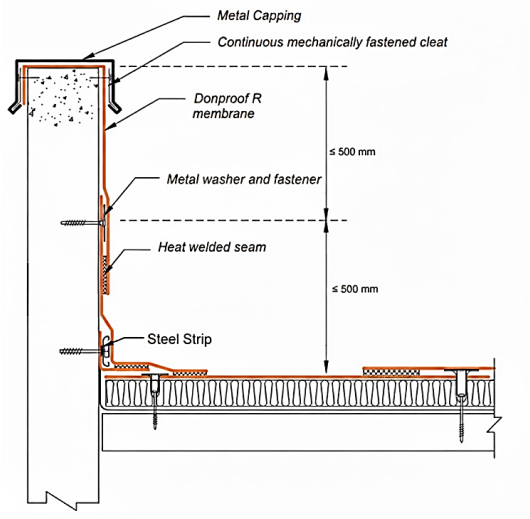


7.2 Flashing to high parapet

7.2.1 For parapets taller than 500 mm, the flashing membrane must be fully bonded to the vertical substrate using contact adhesive and mechanically fixed at the top.

7.2.2 As an alternative to full bonding, the membrane flashing can be loose laid with an intermediate mechanical fastener row. This can be executed either as a two-part covered connection or as a single membrane strip with an intermediate fastener and cover strip.

7.2.3 Max vertical spacing between fastener rows is 500 mm.

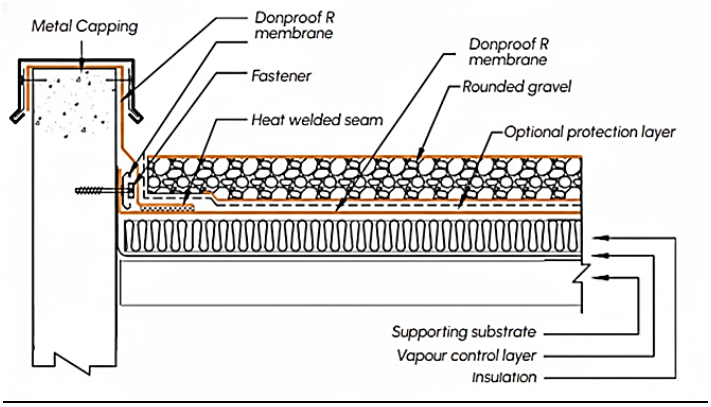


7.3 Parapet flashing on protected constructions

7.3.1 Strong winds can displace gravel ballast (wind scouring), so mechanical fixings are required along the roof perimeter to prevent damage and resist horizontal forces.

7.3.2 For high-rises and/or wind-exposed locations, it is recommended to use concrete slabs in the edge and corner areas.

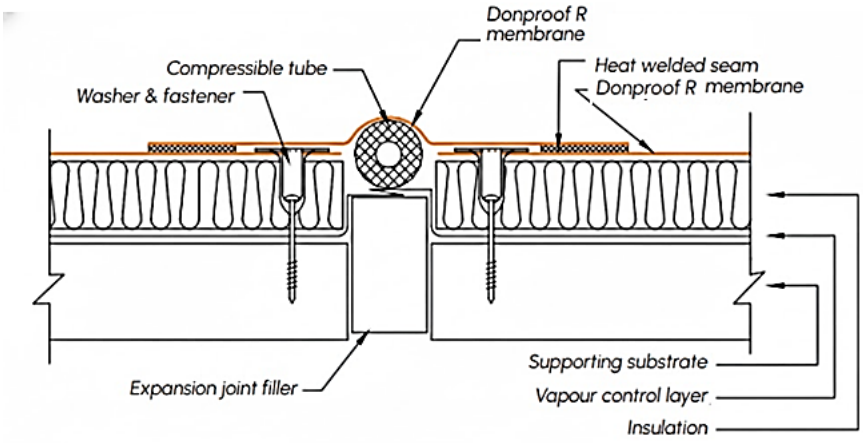
7.3.3 Ballast design must follow Flat Roof guidelines and EN 1991-1-4, along with relevant national standards.



7.4 Expansion joints

7.4.1 The layers in the structure of the roof must have joints wherever there are joints in the underlying supporting structure of the building.

7.4.2 Structural and expansion joints should normally be designed as high points. They must not run through openings in the roof and must not be closer than 500 mm to such points.



7.5 Flashing to rainwater drainage outlets

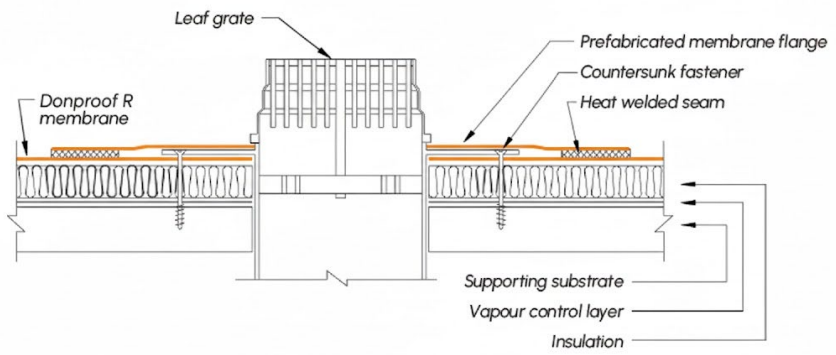
7.5.1 It is recommended to use prefabricated drains with factory-welded membrane collars or drains equipped with clamping rings. Drains should be positioned at least 300 mm away from walls, parapets, joints, or other structural elements.

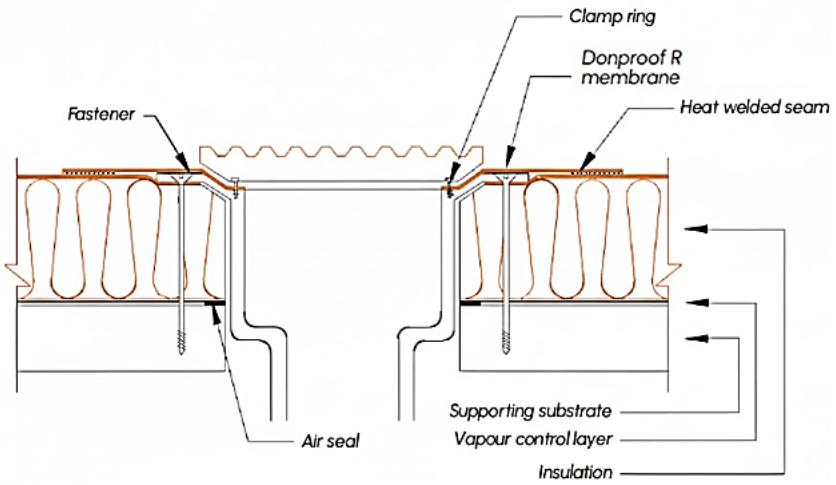
7.5.2 Each section separated by an expansion joint must have its own dedicated drain. Always follow local or national regulations.

7.5.3 Where roof outlets include clamping flanges and sealing gaskets, the membrane can be securely clamped between the top and bottom gaskets.

7.5.4 The drain must be installed at the lowest point of the roof, and if needed, the thermal insulation should be recessed to form a shallow sump for proper drainage.

7.5.5 The drainage system must be sized according to national requirements, and provisions for emergency overflows must also be included in the design.





7.6 **Flashing to pipe penetrations**

7.6.1 Pipe penetrations are normally sealed using prefabricated pipe details from our accessories range - or from a flange approx. 500 x 500 mm and a collar made from homogeneous membrane.

7.6.2 Four fasteners must be installed in the field membrane around the base of the pipe. In the middle of the flange a hole must be cut.

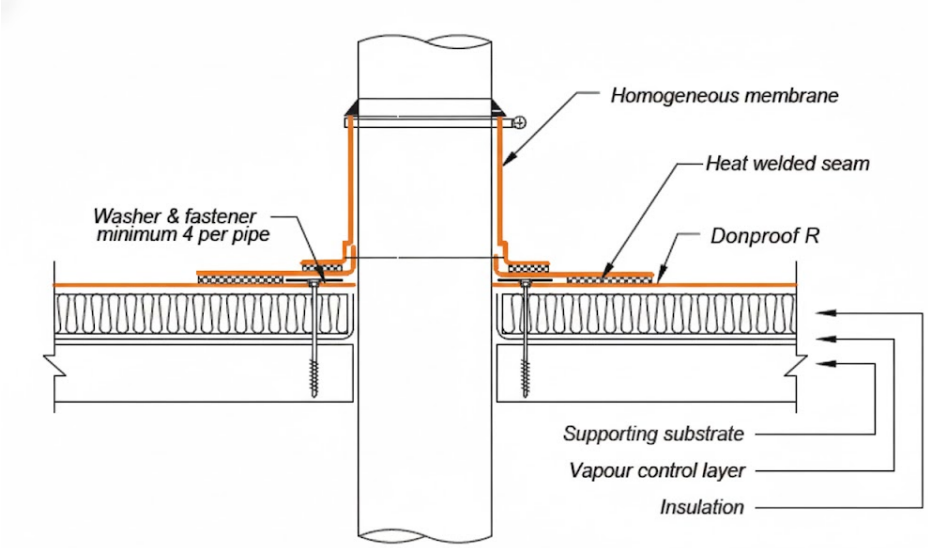
7.6.3 The hole should be approx. 60 mm smaller than the pipe diameter.

7.6.4 Using a hot air hand gun/welder, the flange is heated to make it pliable and plastic so that it easily can be pulled over the protruding pipe.

7.6.5 In some cases, it is easier to make this shape adjustment on a loose pipe with the same diameter.

7.6.6 Because of the smaller diameter in the membrane collar, the collar material stands up a few centimeters around the pipe. The collar must be welded to the field membrane.

7.6.7 The flashing strip can now be installed and hot air welded to the up standing material from the collar.



8.0 Cleaning

8.1 Tools and equipment can be cleaned with water.

Section C: Cautions

Health and safety

- Apply in well-ventilated areas. In closed areas use force ventilation and wear approved respirators when necessary.
- **DonProof R** should not come into contact with skin and eyes. In case of contact with eyes wash immediately with plenty of water and seek medical advice promptly.
- Vapor and atomized liquids are harmful.

Fire:

DonProof R is nonflammable.

For further information on refer to the Material Safety Data Sheet.

Section D: Approval and Variations

This method statement is offered by DCP as a 'standard proposal' for the application of **DonProof R**. It remains the responsibility of the Engineer to determine the correct method for any given application. Where alternative methods are to be used, these must be submitted to DCP for approval, in writing, prior to commencement of any work. DCP will not accept responsibility or liability for variations to the above method statement under any other condition.